PMSM Modelling in Motor-CAD and Reduced Order Model Extraction

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Mathworks-Synopsys Video Series: Video 1

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Agenda

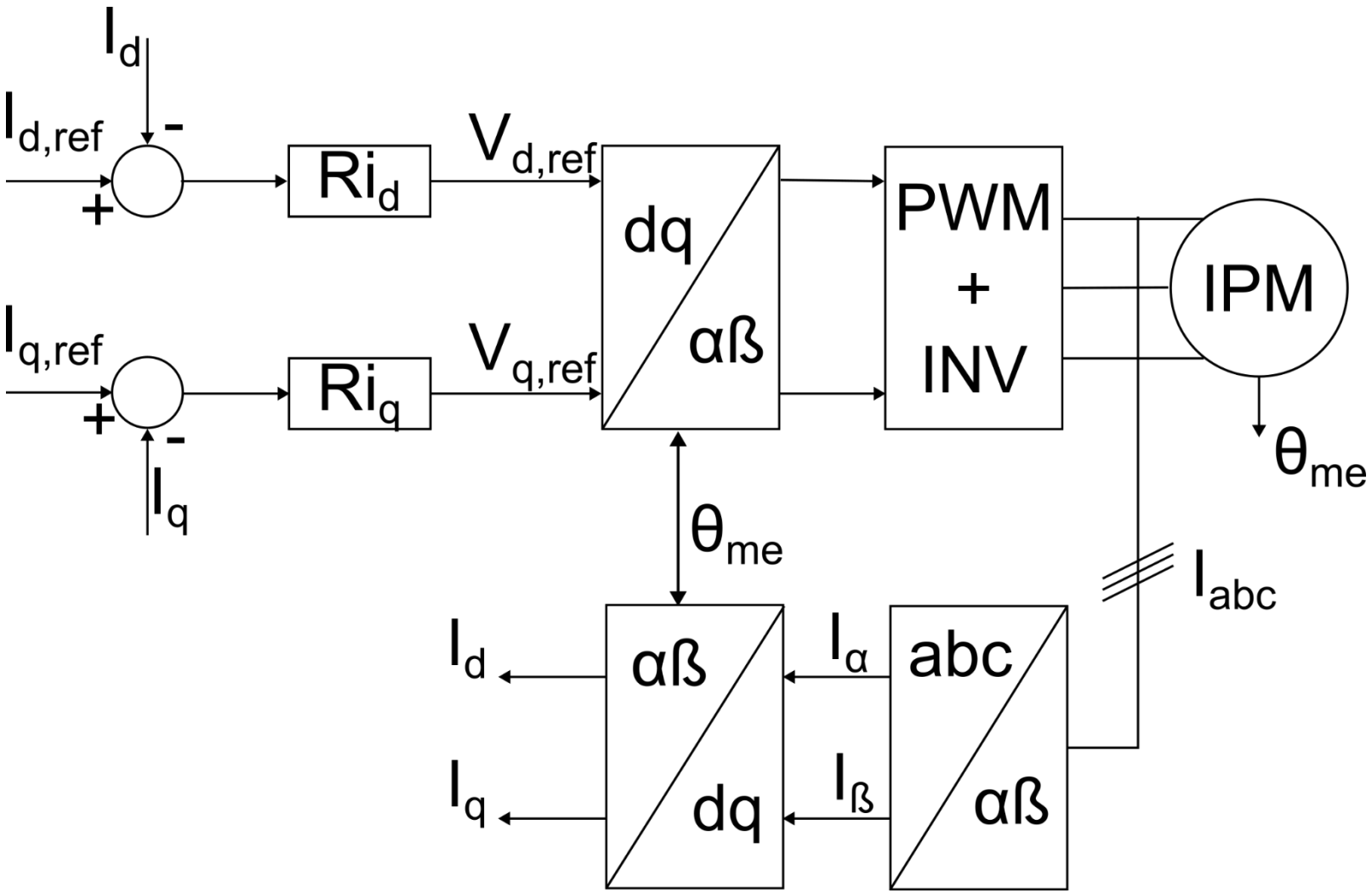
- Introduction and motivations.
- What does it mean extracting a Reduced Order Model of an electrical machine.
- Model preparation and electromagnetic analysis in Motor-CAD.
- ROM extraction in Motor-CAD.

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Introduction

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Motor Control Scheme



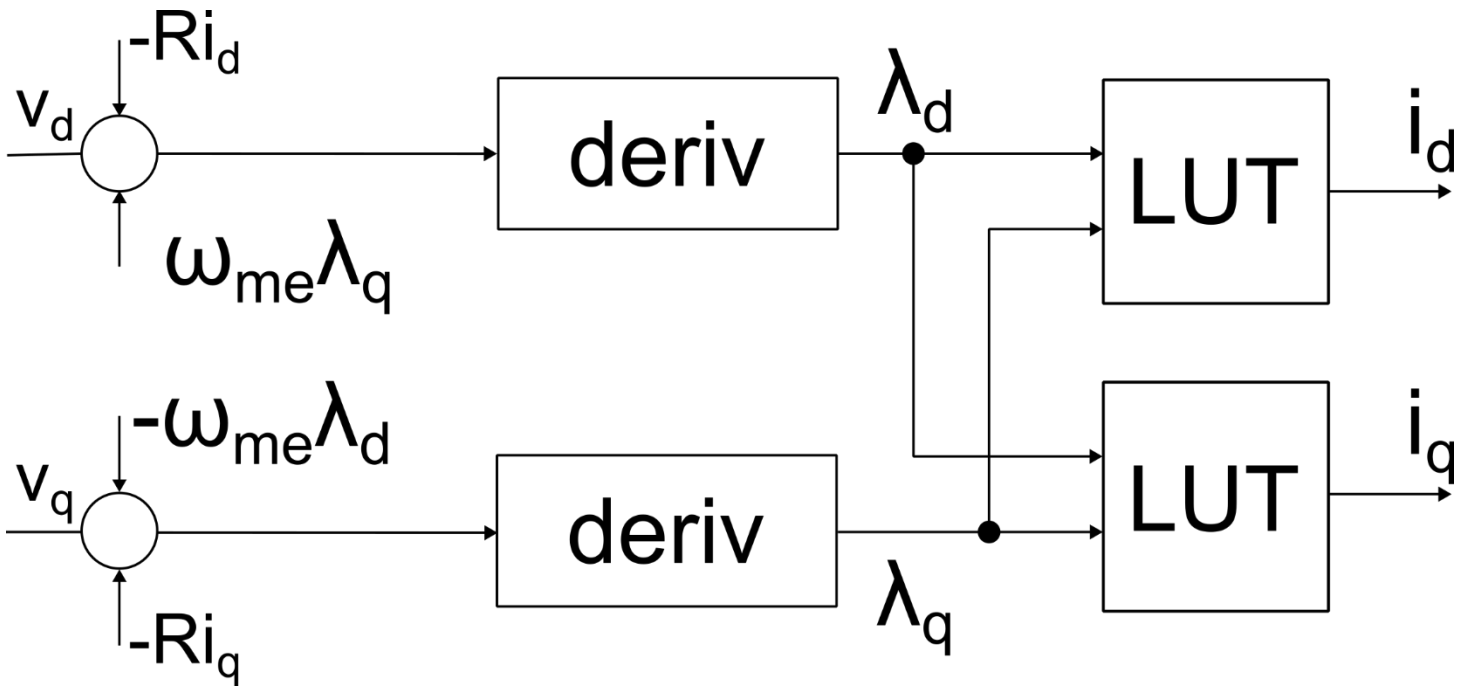
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Motor Model

- The LUT are tables connecting the machine current and flux linkage.

$$v_{\mathrm{d}} = Ri_{\mathrm{d}} + \frac{d\lambda_{\mathrm{d}}}{dt} - \omega_{\mathrm{me}}\lambda_{\mathrm{q}}$$
$$v_{\mathrm{q}} = Ri_{\mathrm{d}} + \frac{d\lambda_{\mathrm{q}}}{dt} + \omega_{\mathrm{me}}\lambda_{\mathrm{q}}$$



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Reduced Order Model of an Electrical Machine

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Reduced Order Model of an Electrical Machine (1)

- An electrical machine is a complex problem that needs to be solved with the help of the FEA.
- The iron saturation affects the overall performance but also any type of ripple of the main quantities and closely depends on the geometry.
- The correct evaluation of the field in every part of the machine is necessary to compute the solid and iron losses.
- A high-fidelity machine model is necessary also for the correct definition of the control parameters.
- Not only the saturation, but also the flux ripple and losses require correct tuning for the drive dynamic and the optimal operating point selection.

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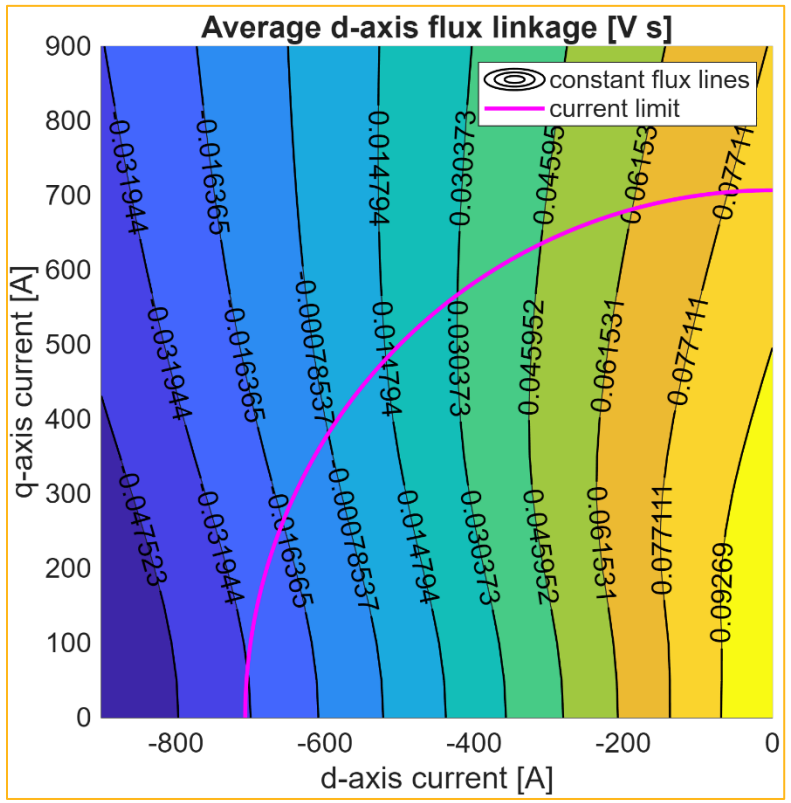
Reduced Order Model of an Electrical Machine (2)

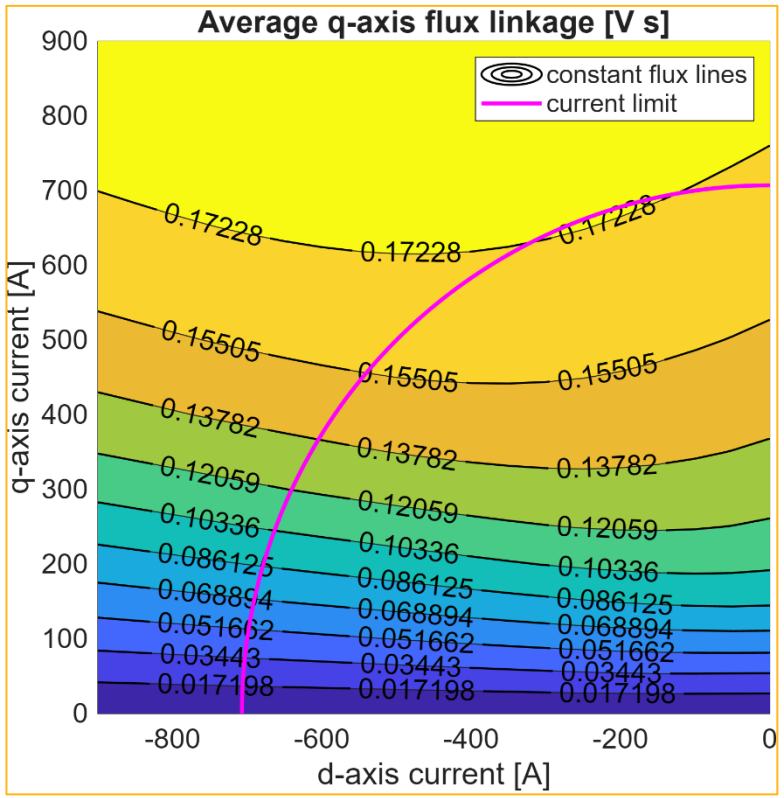
- From the machine modelling point of view, the idea is to produce matrices where the flux, torque and losses are linked to a specific stator current and rotor position combination.
- For each (I_d, I_q) operating point the machine flux, torque and losses are computed varying the rotor position, at fixed speed.
- The iterative procedure is automatically done with a specific Motor-CAD tool, which produces a .mat file with all the matrices necessary for the detailed reduced order model of the machine.

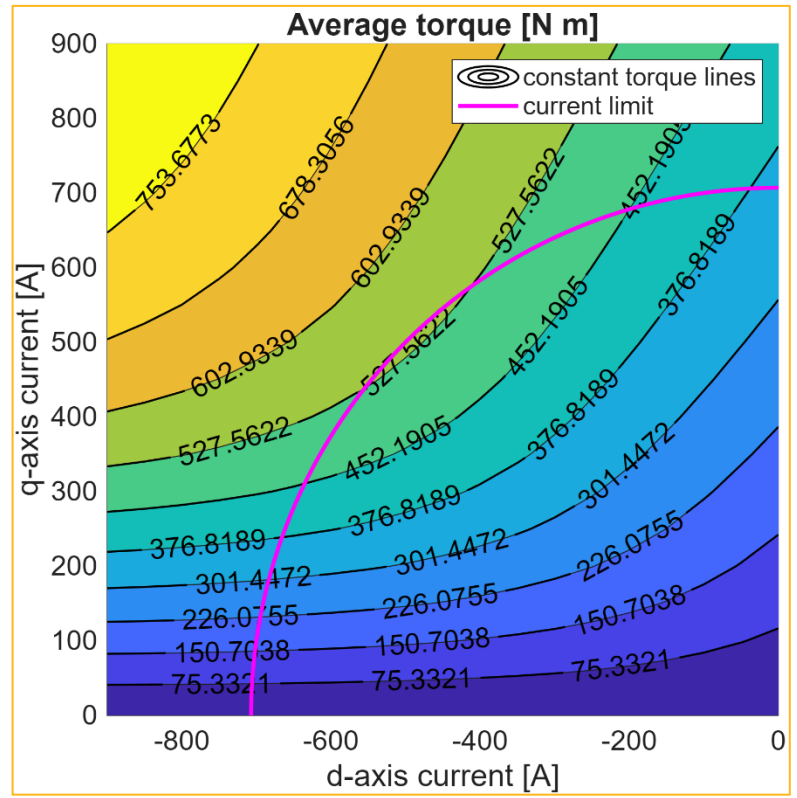
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Reduced Order Model of an Electrical Machine (4)

- Flux linkage and torque maps.



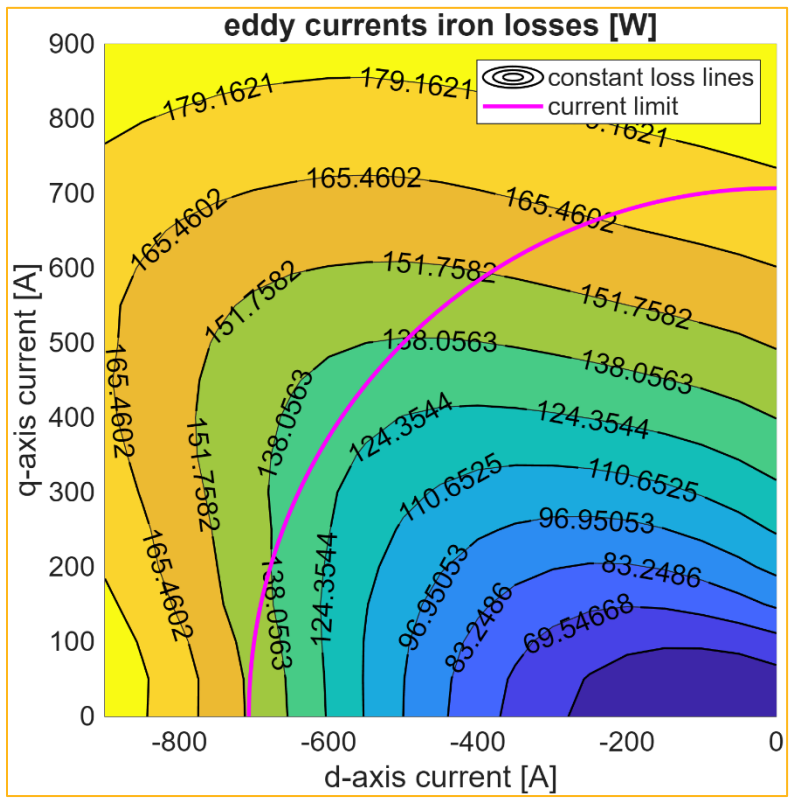


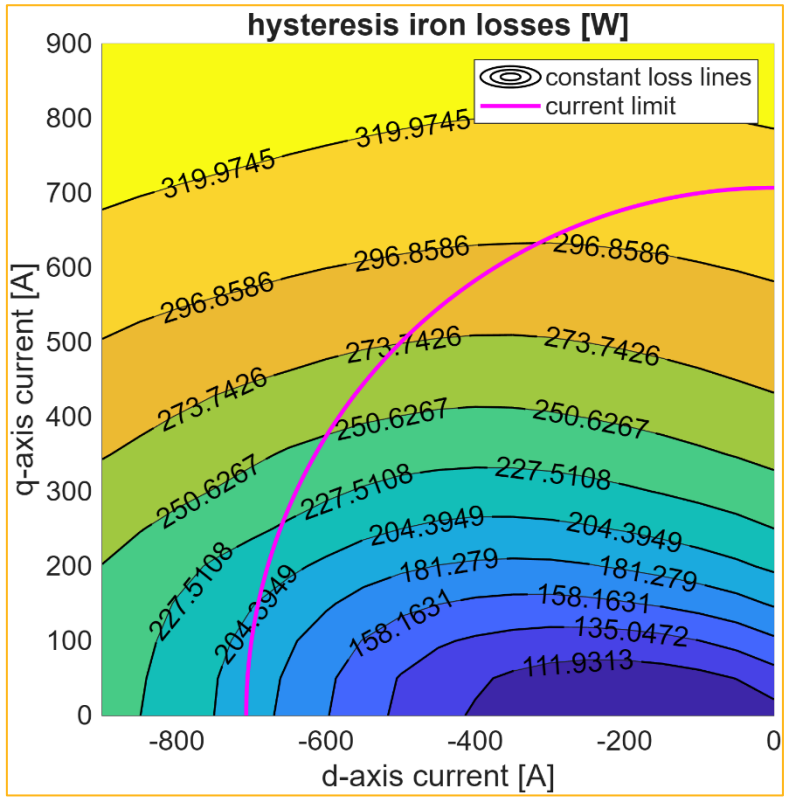


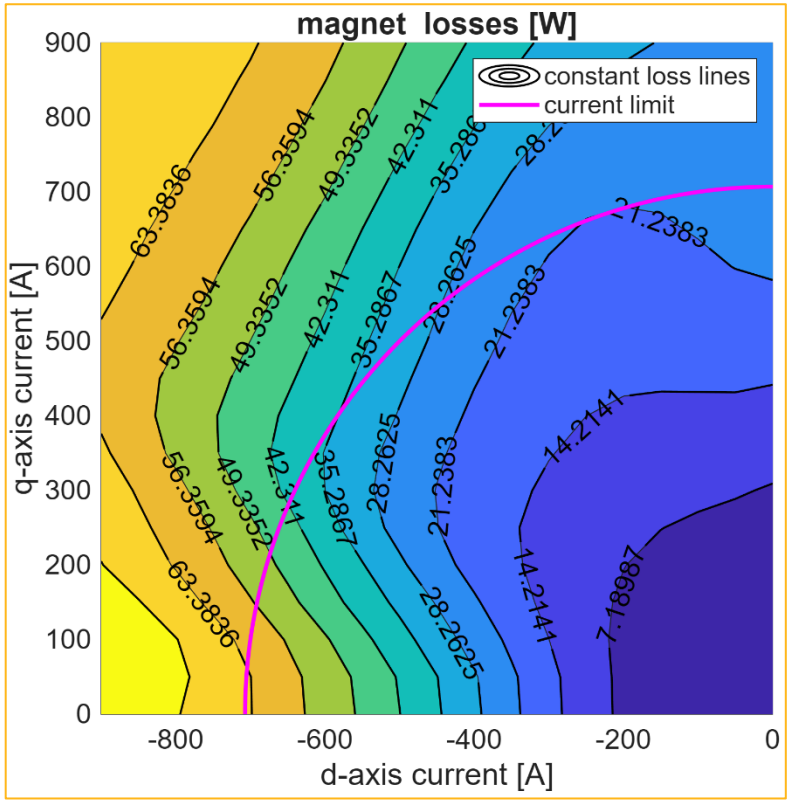
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Reduced Order Model of an Electrical Machine (5)

- Losses maps.





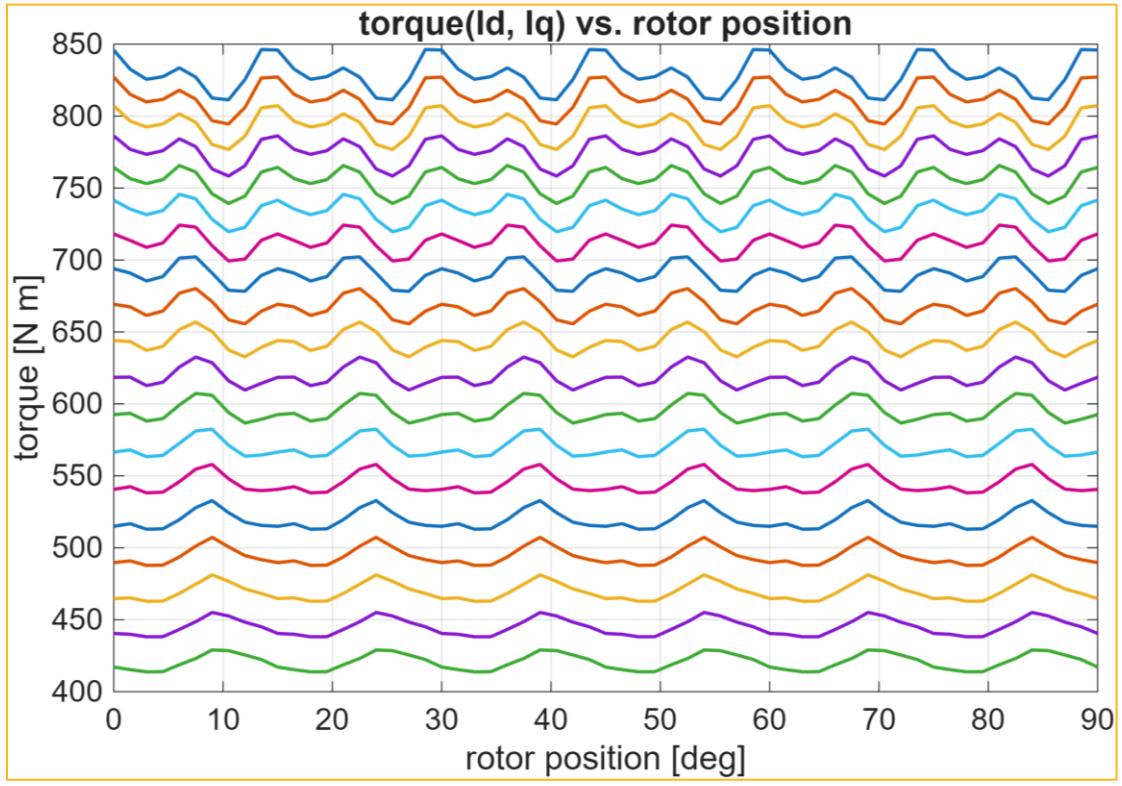


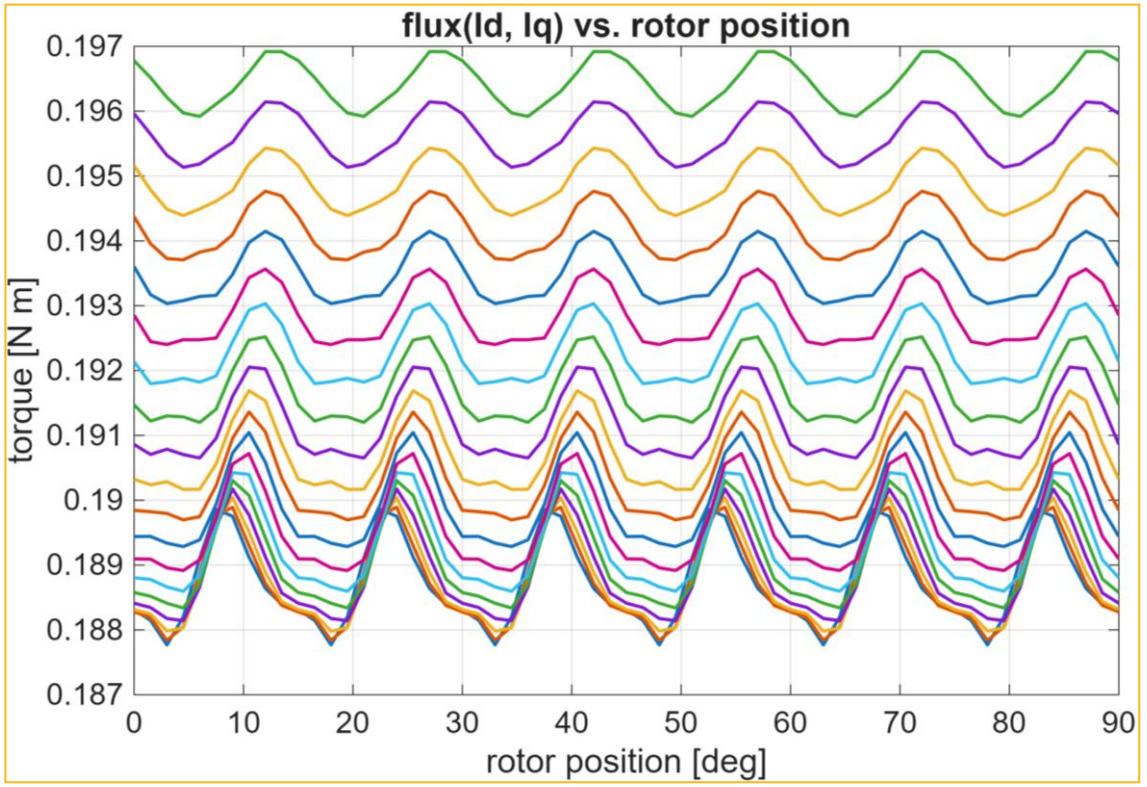
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Reduced Order Model of an Electrical Machine (6)

- Torque and flux linkage vs. rotor position.





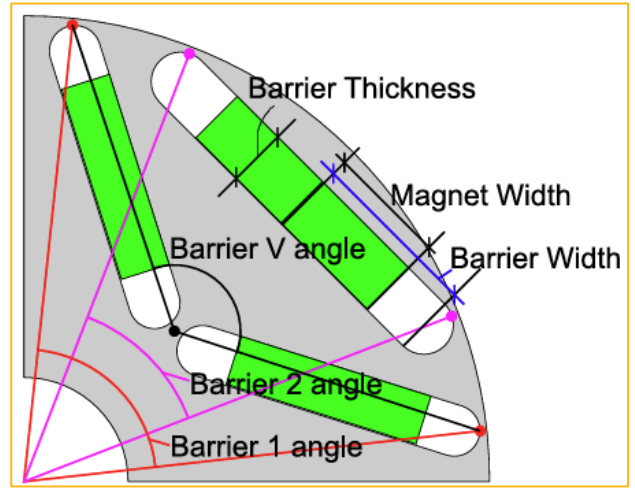
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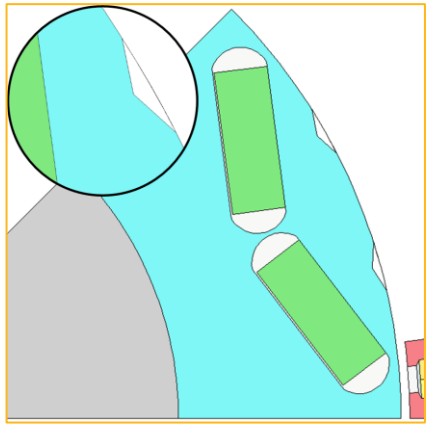
Machine Model Preparation and Electromagnetic Analysis in Motor-CAD

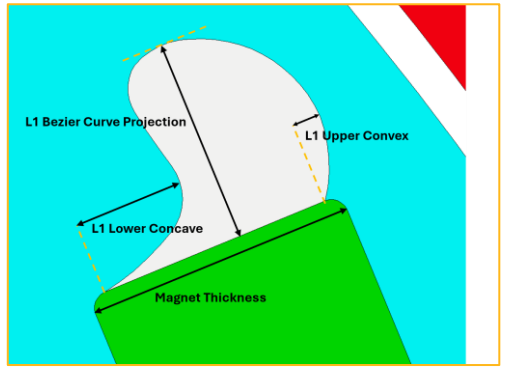
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Geometry Definition (1)

- The geometry definiton can be done in several ways in Motor-CAD.
- Motor-CAD templates define an electric machine through parameters rather than individual sketch entities.
- Standard templates provide the machine topology. Adaptive Templates let you extend that architecture with your own geometry intelligence.
- Custom geometrical features can be added with the adaptive template like stator and rotor notches, holes, special barriers geometries.



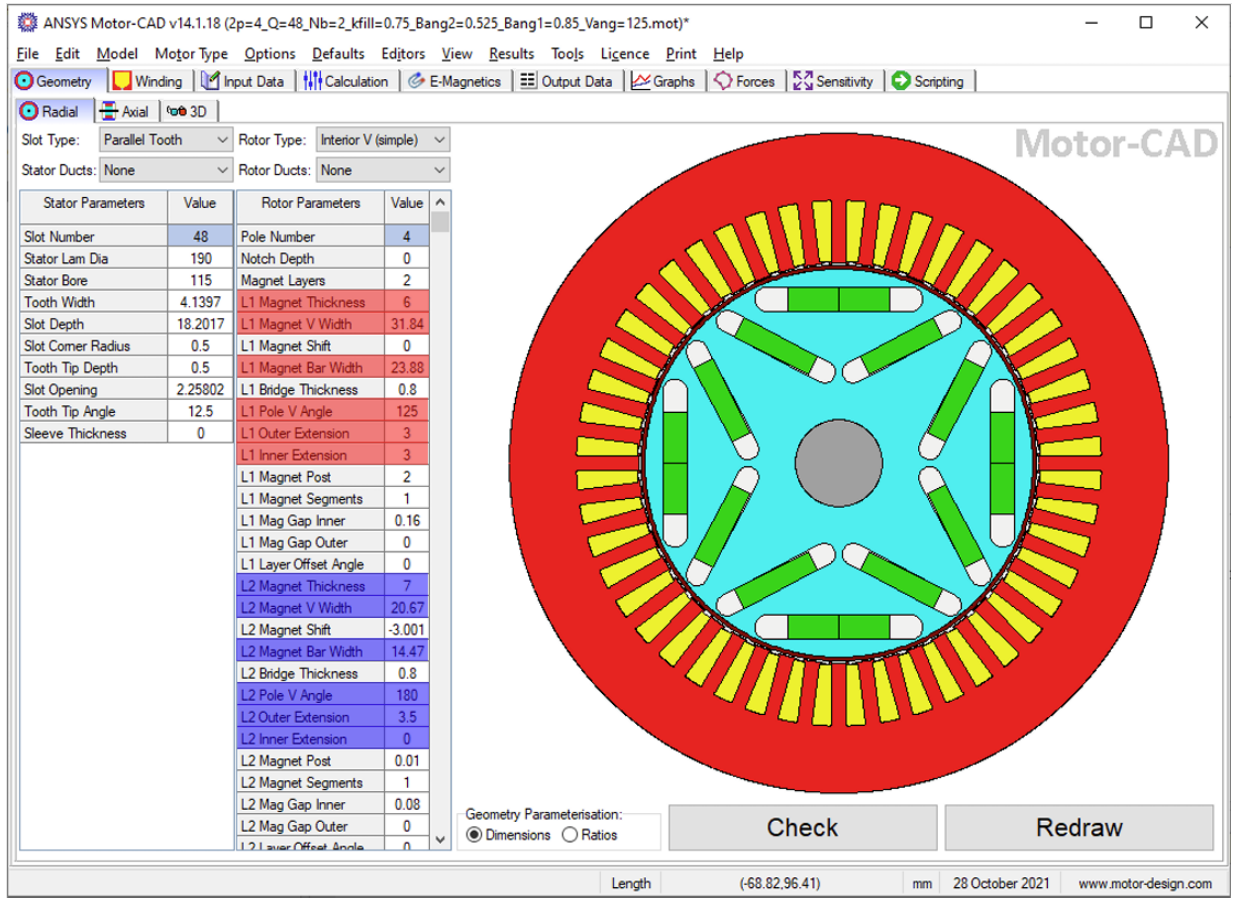


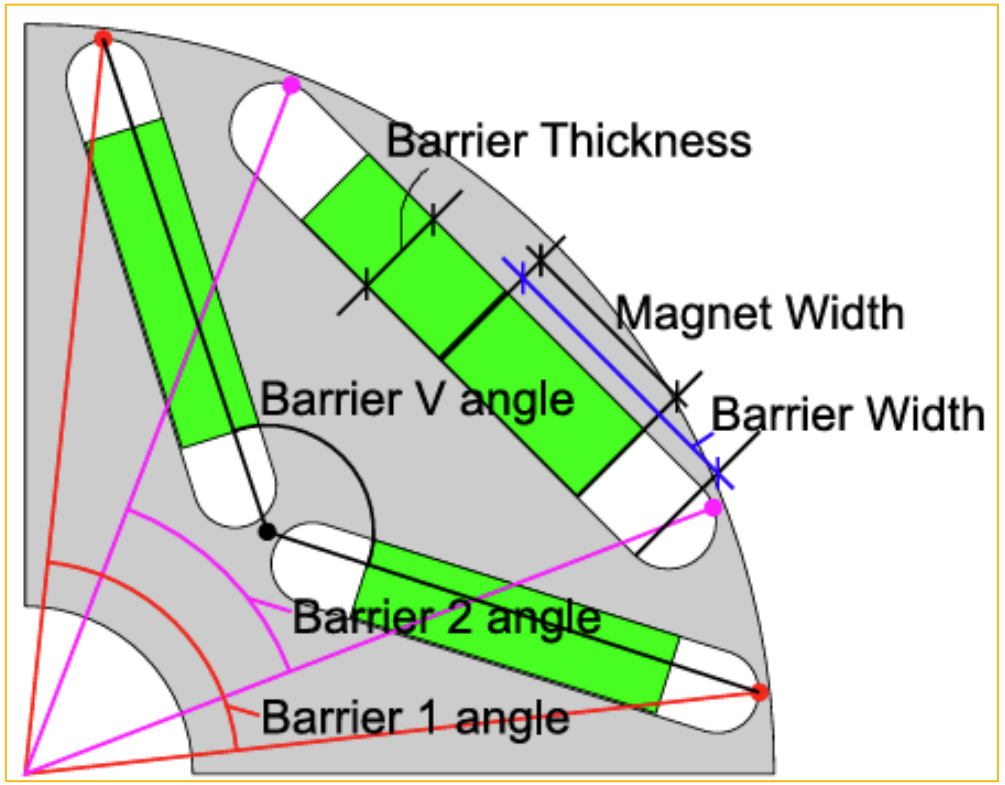


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Geometry Definition (2)

- The template geometry is fully controlled by the dedicated parameters.

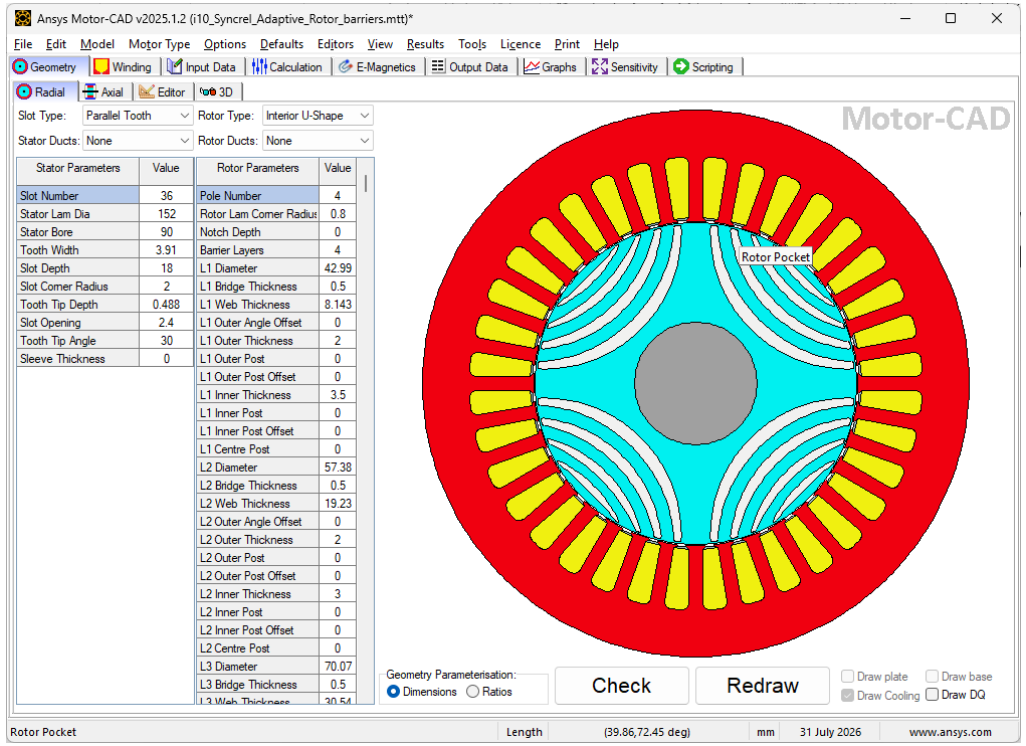


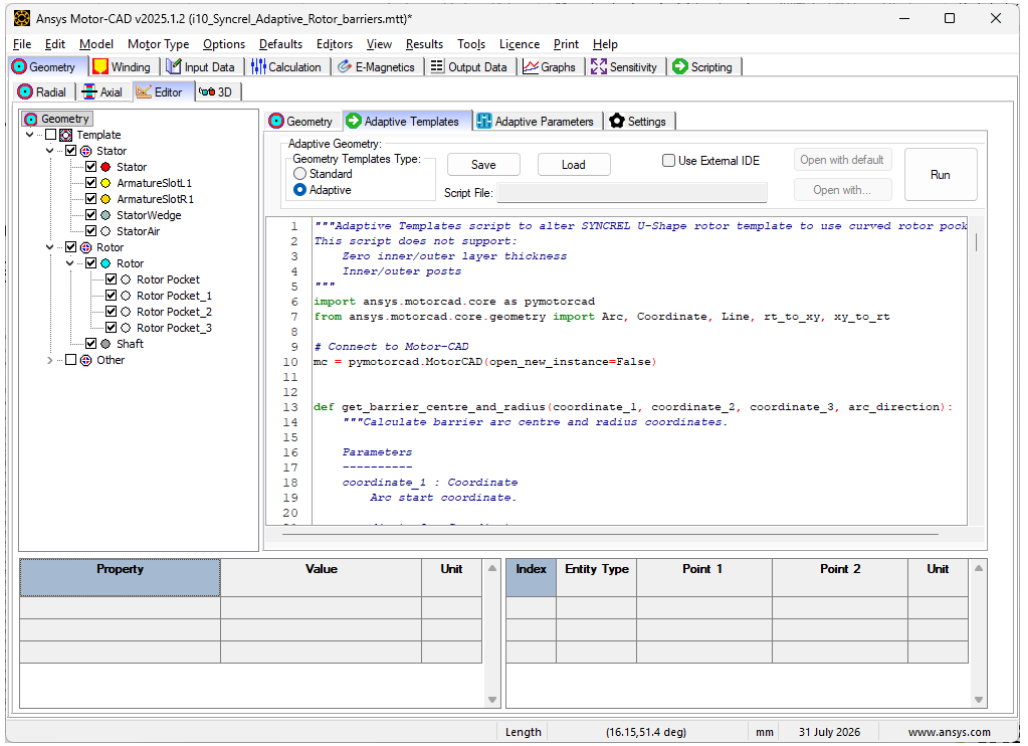


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Geometry Definition (3)

- Adaptive Templates combine Motor-CAD's template-based workflow with Python scripting, enabling users to extend standard machine templates with fully customizable, parameterized geometry features.

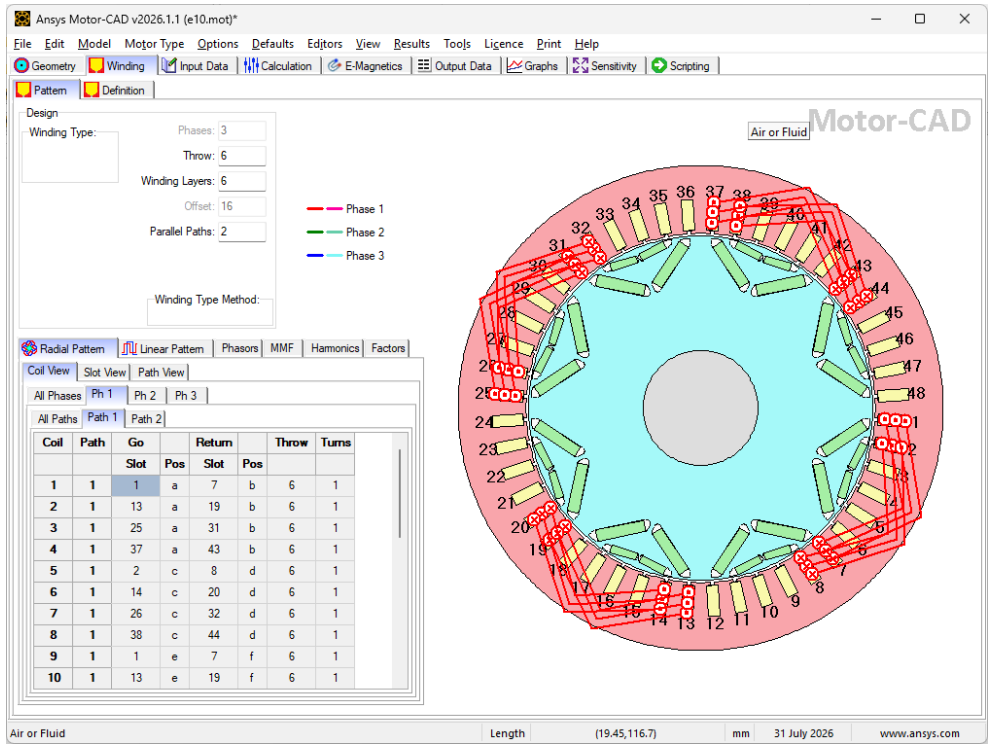


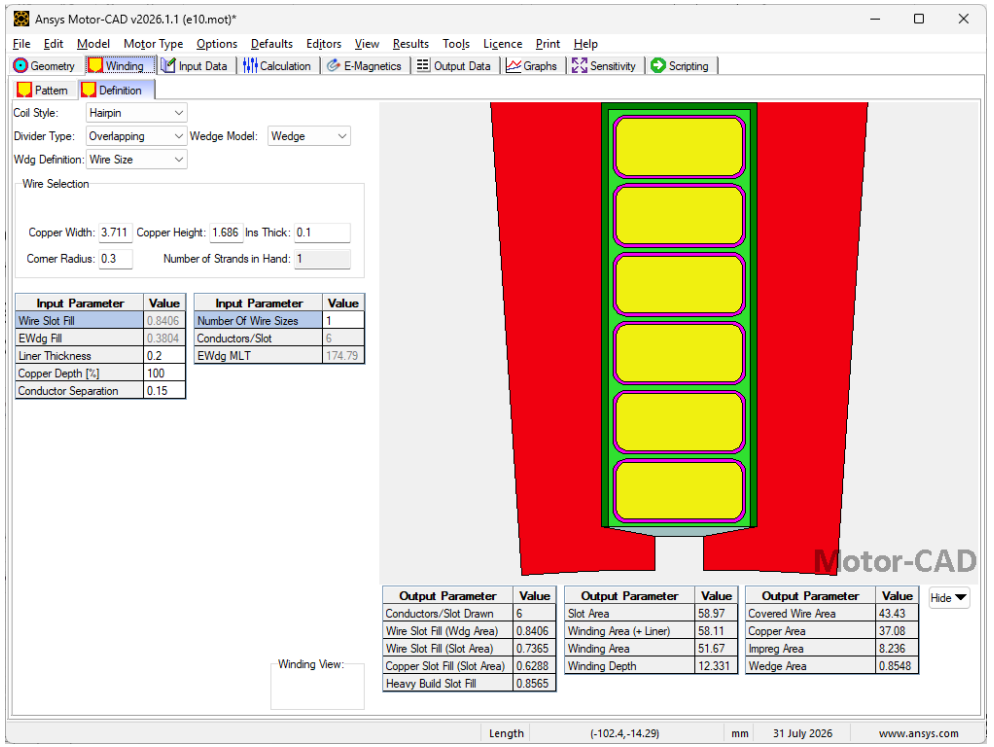


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Winding Definition

- From conventional wire windings to advanced hairpin topologies, Motor-CAD automatically defines winding patterns, parallel paths, and end-winding geometry, allowing engineers to focus on machine performance rather than manual winding construction.

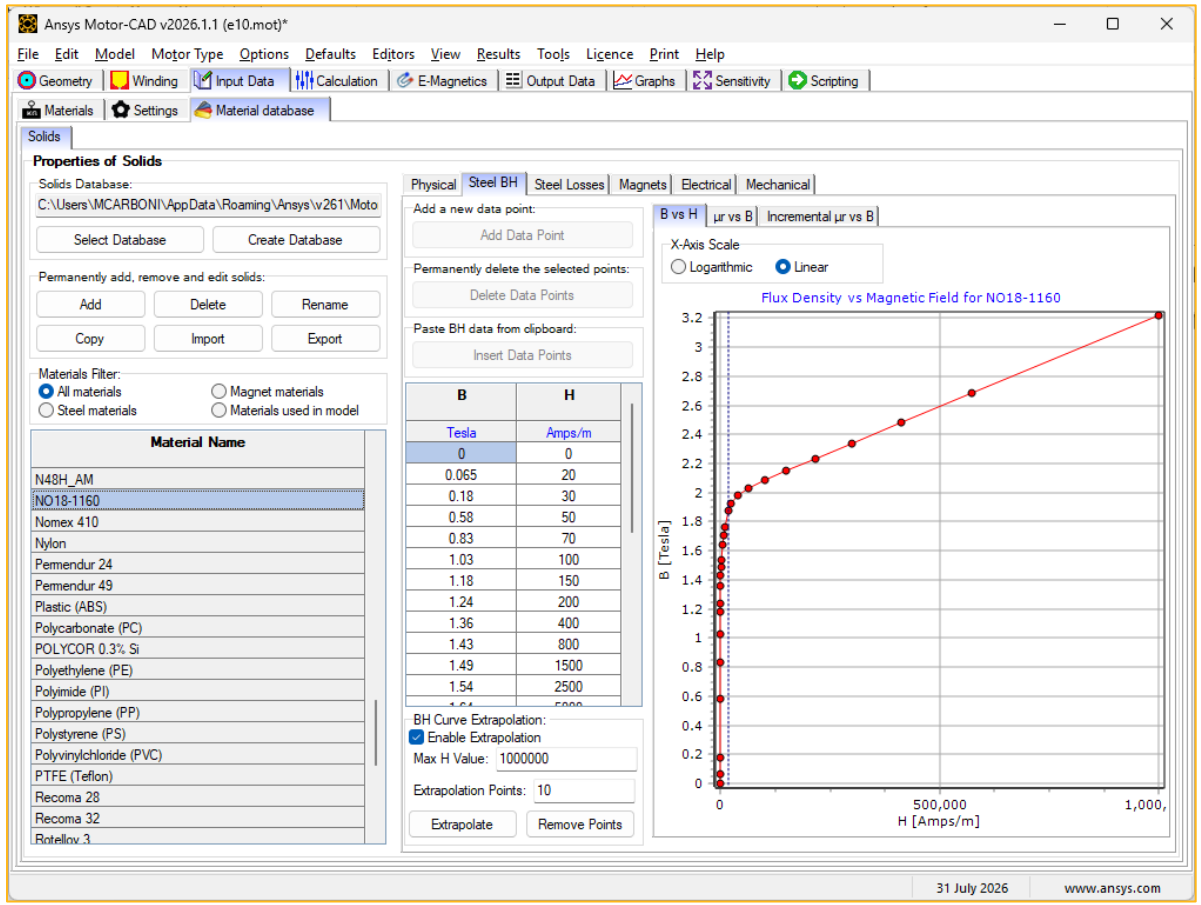


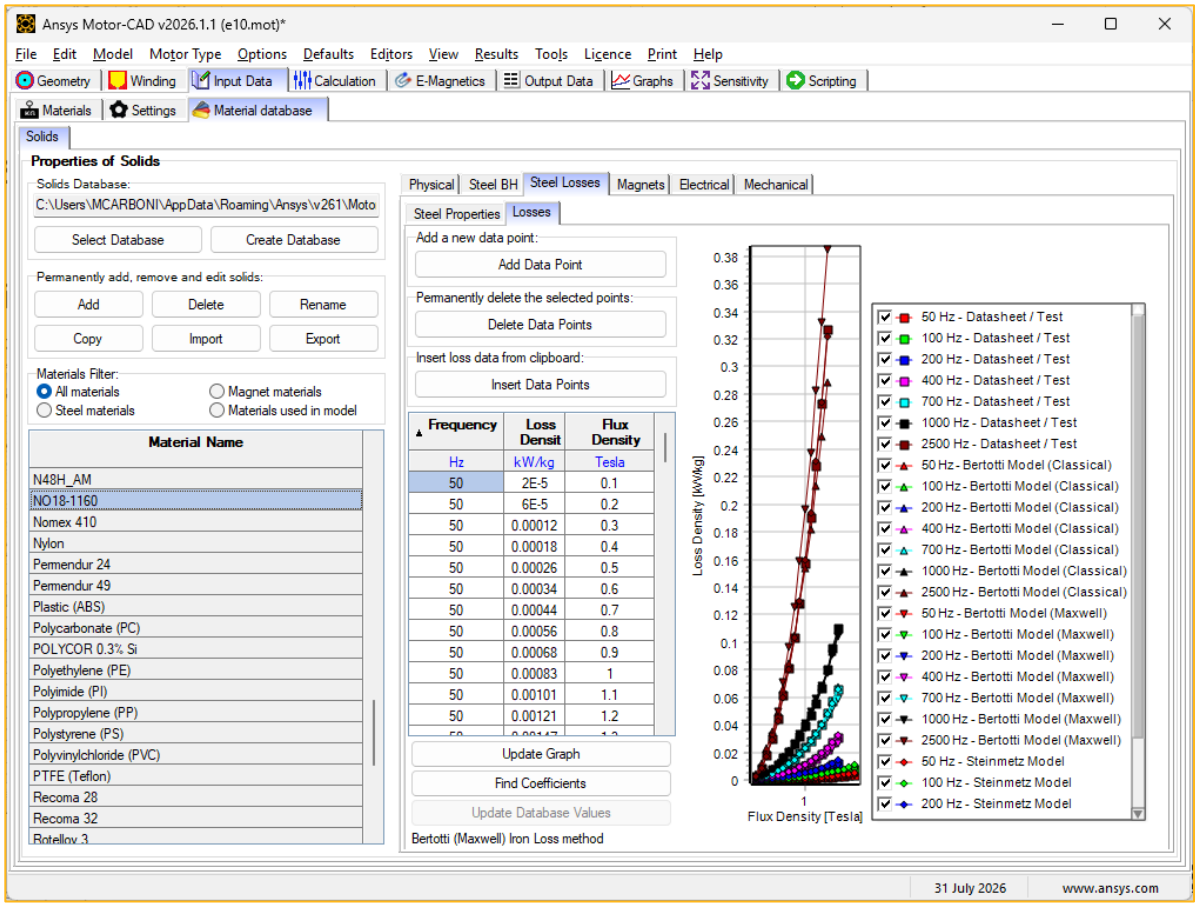


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Materials (1)

- Defines the magnetic B-H behavior and core-loss properties of the stator and rotor laminations used for electromagnetic FEA and performance prediction

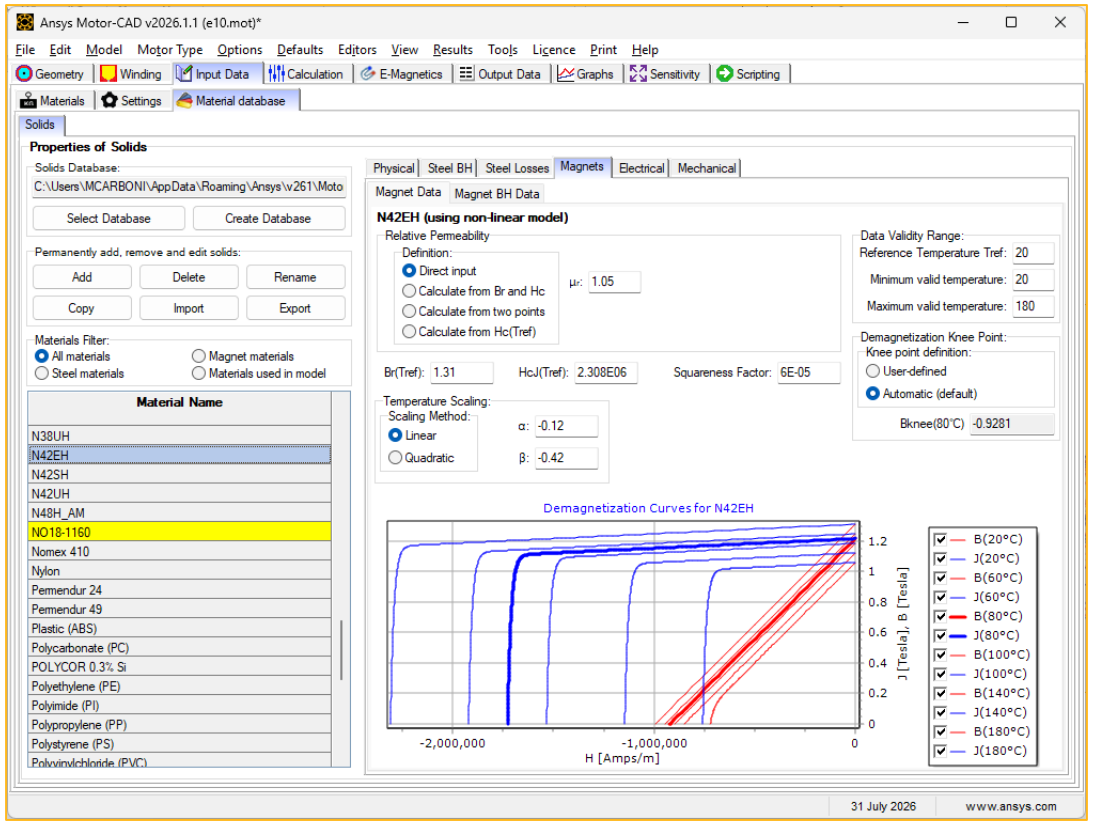


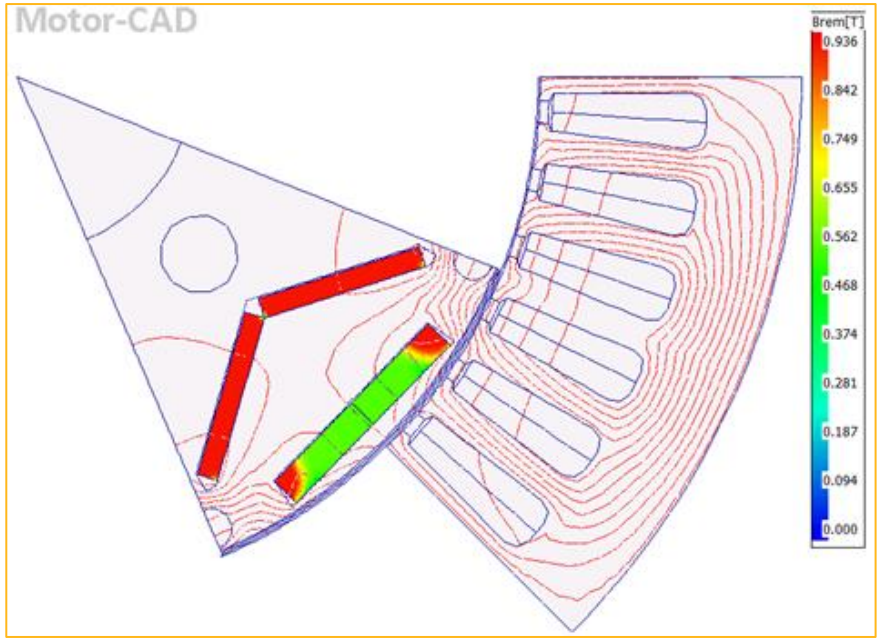


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Materials (2)

- Non-linear magnet material definition for correct prediction of the remanence and Back-EMF with demagnetization.
- Magnet losses are predicted in the transient simulation.

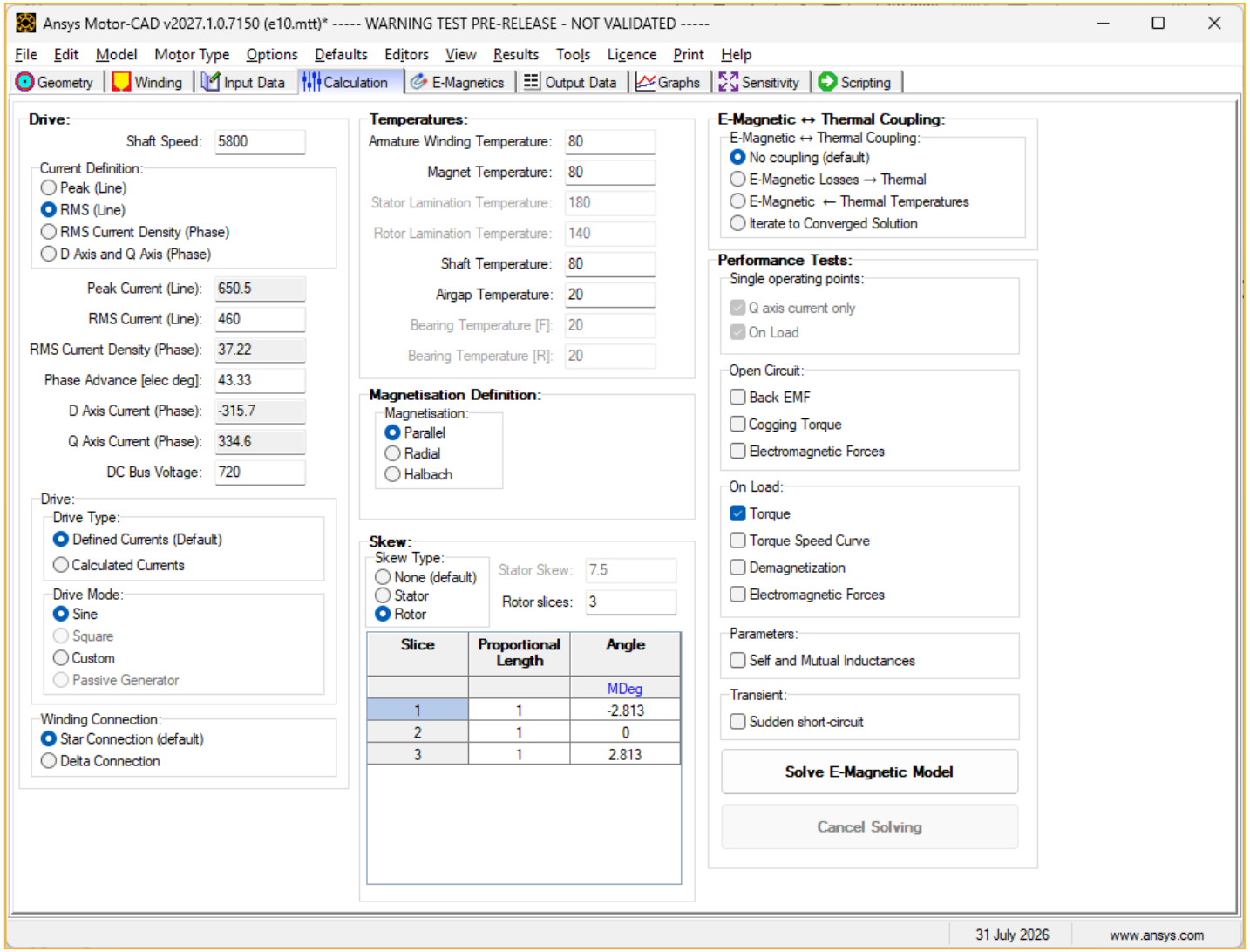




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Operating Point Calculation Settings

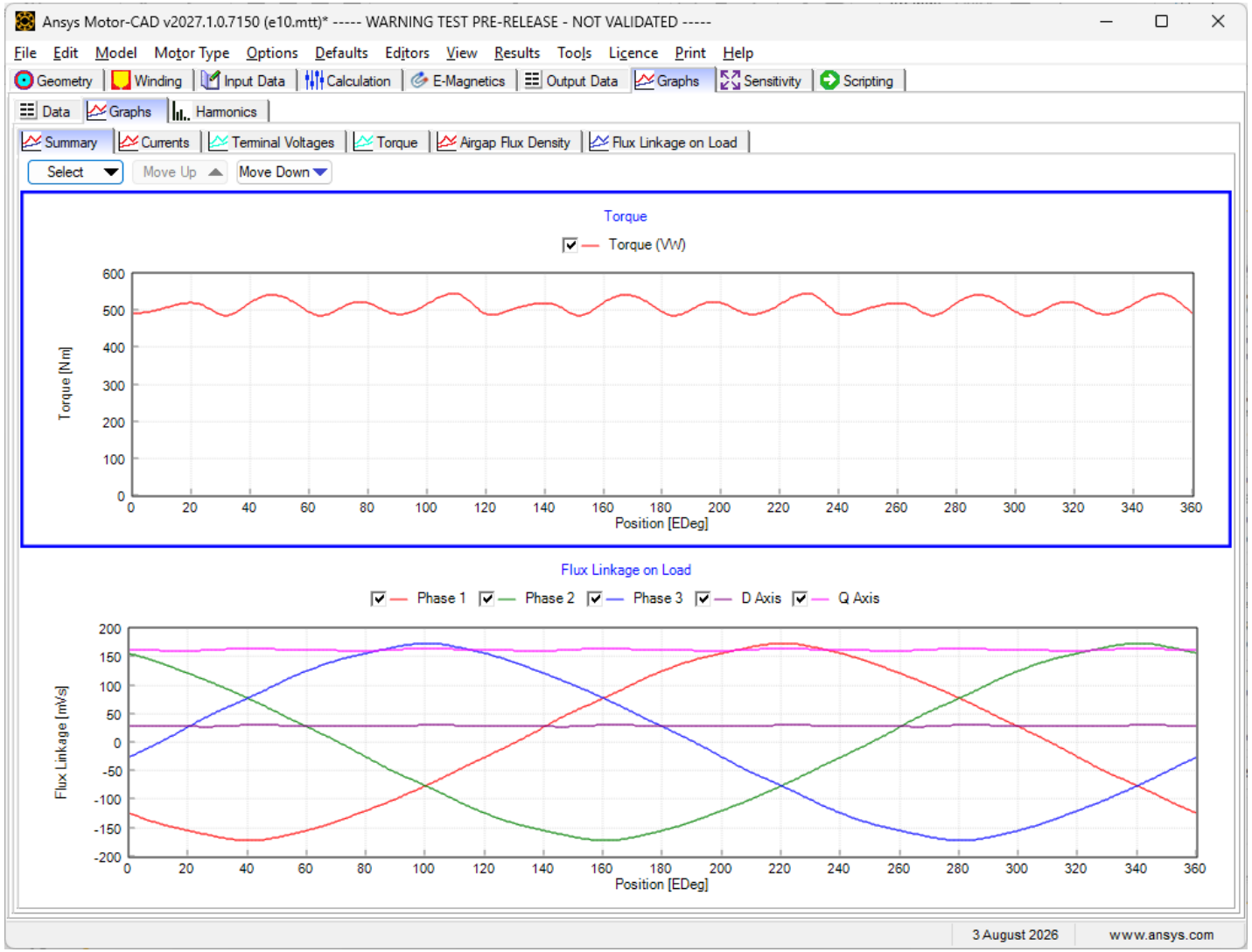
- Defines the machine operating point and simulation conditions, including current, speed, temperatures, and drive settings.
- Current amplitude and angle.
- Stator or rotor skewing.



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Torque and Flux Linkage Calculation

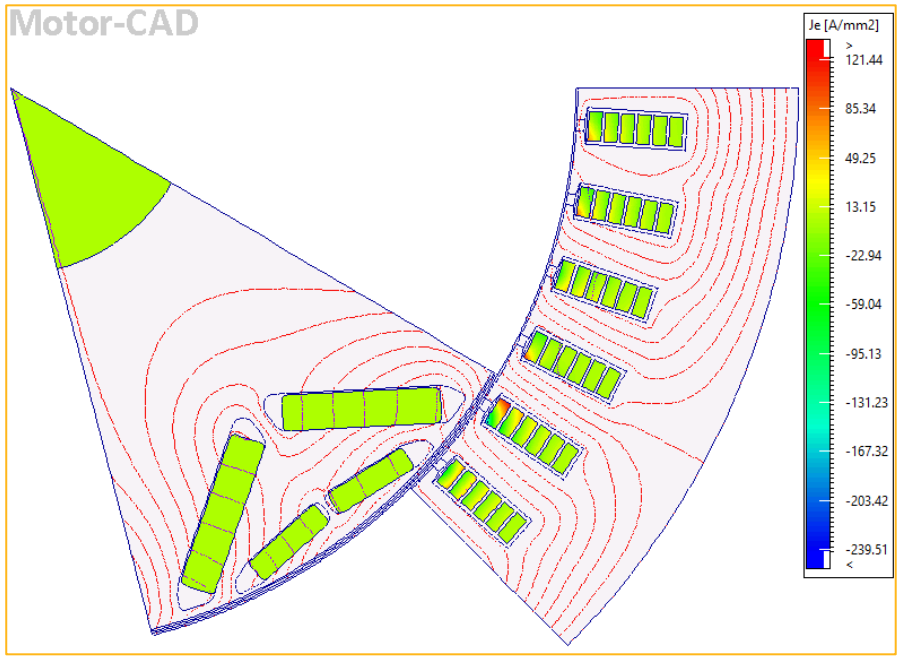
- Torque and flux linkage are computed varying the rotor position.
- This analysis is repeated for each operating point in the ROM calculation.

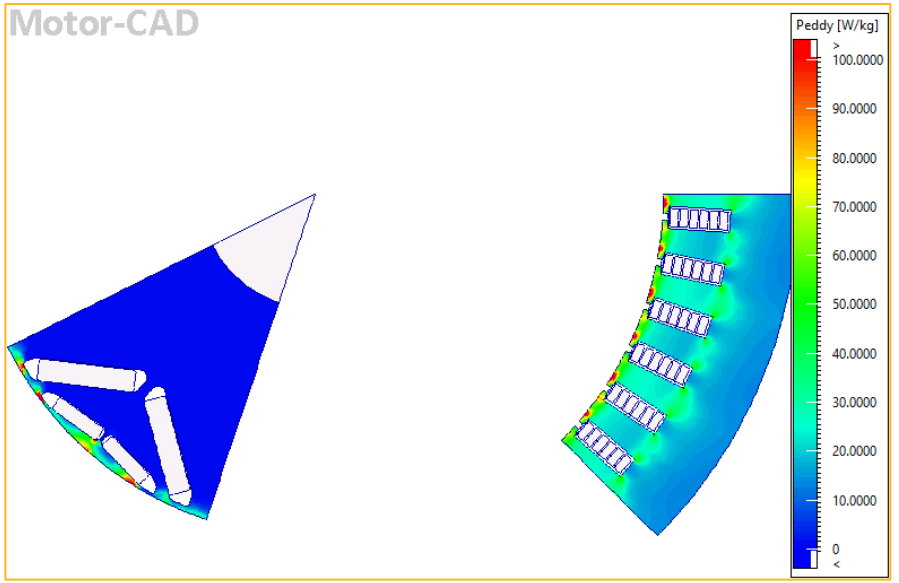


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Losses Calculation

- Iron loss calculation method. Bertotti is the most adopted, since it better fits the transient analysis.
- AC losses calculation type:
  - Hybrid method for a quicker loss calculation. Applies the analytical model using the slot B field from the transient simulation.
  - Full FEA method. All the conductors are modelled in the FEA. Best accuracy but slower simulation.





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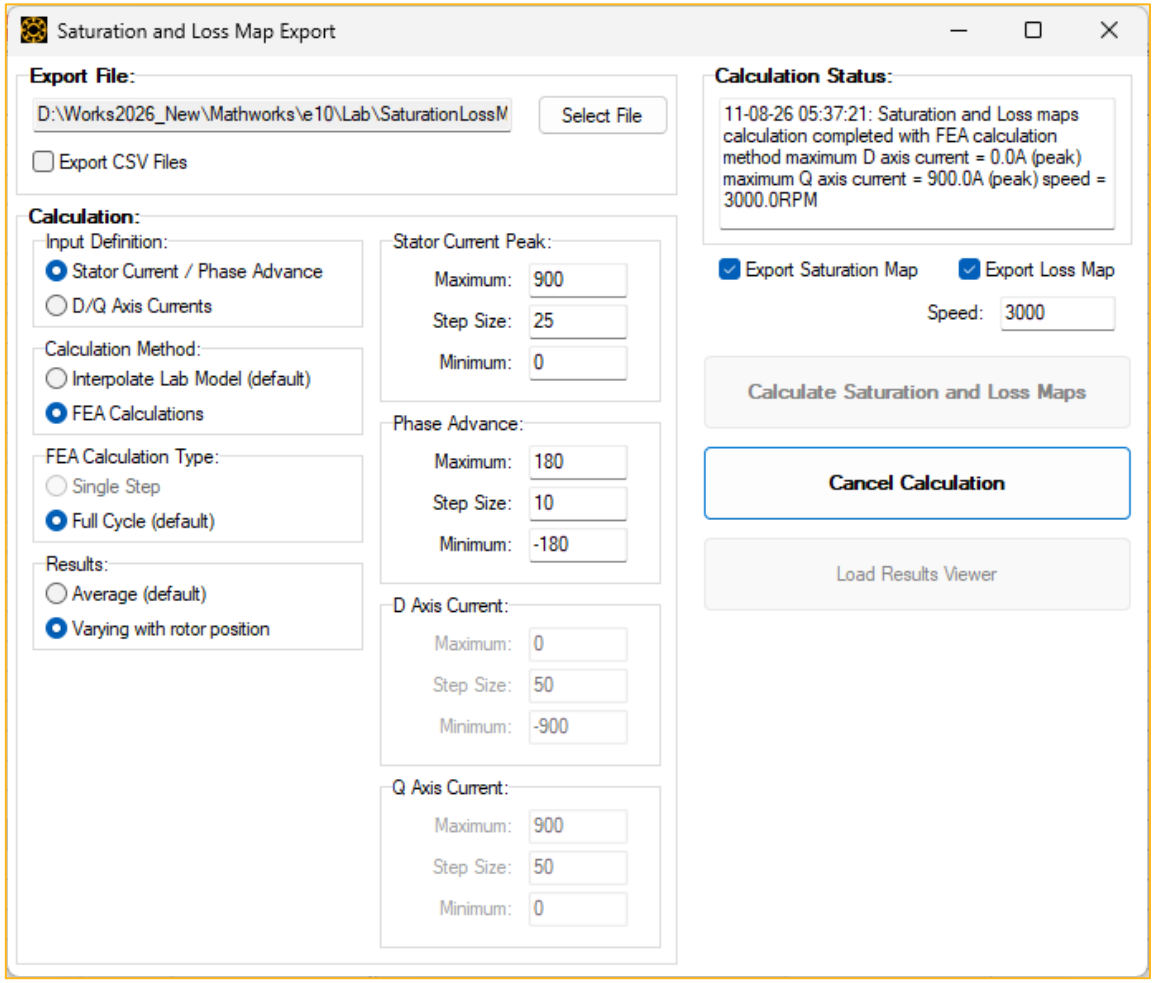
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ROM Extraction With Motor-CAD

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Saturation and Loss Map Export (1)

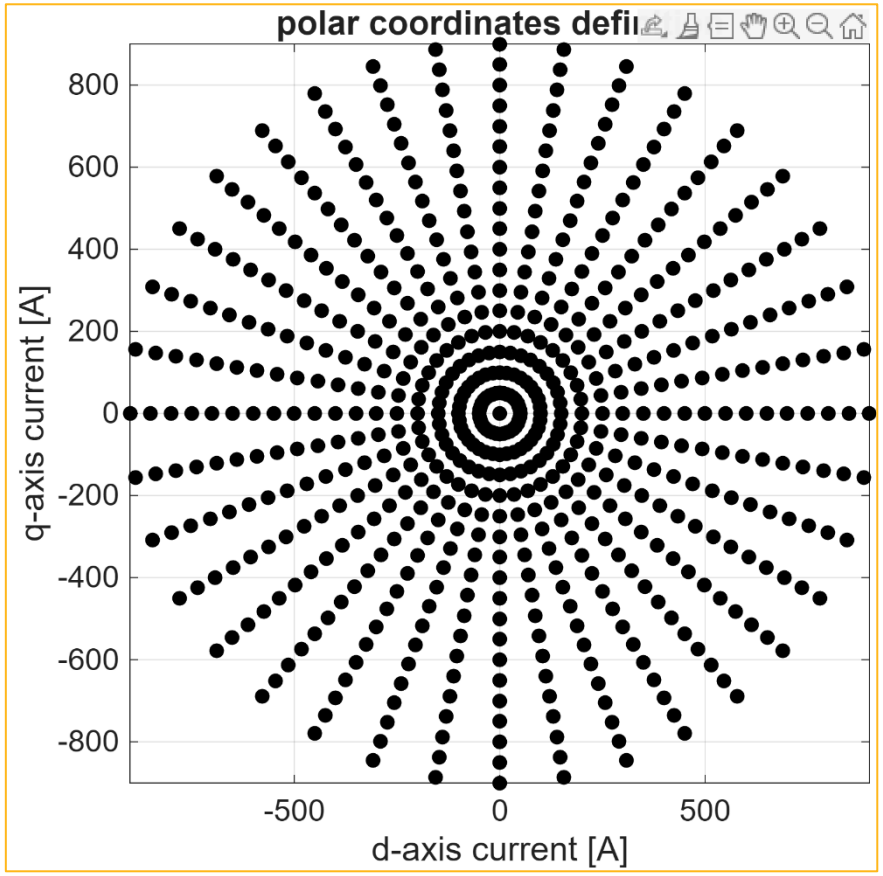
- A guided UI that allows users to configure operating-point sweeps, generate saturation and loss maps, visualize the resulting machine characteristics, and export them for system simulation and optimization workflows.
- Simulink can take them both (average and full cycle).
- Interpolate lab for quick calculation.

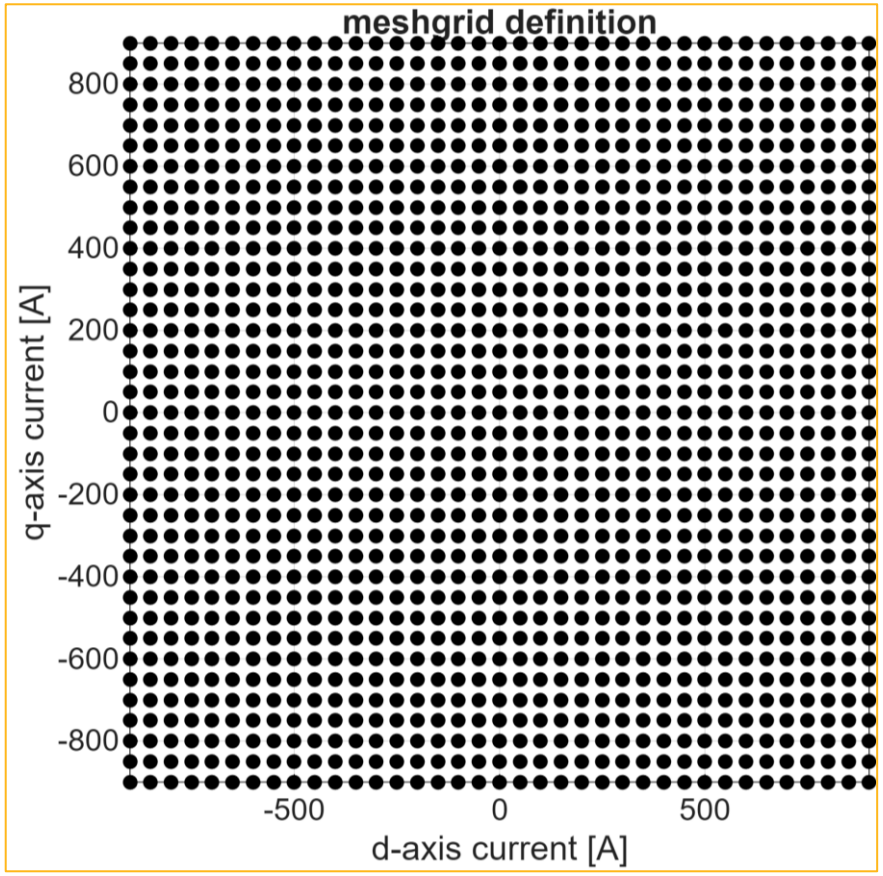


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Saturation and Loss Map Export (2)

- Different definition of the operating points.



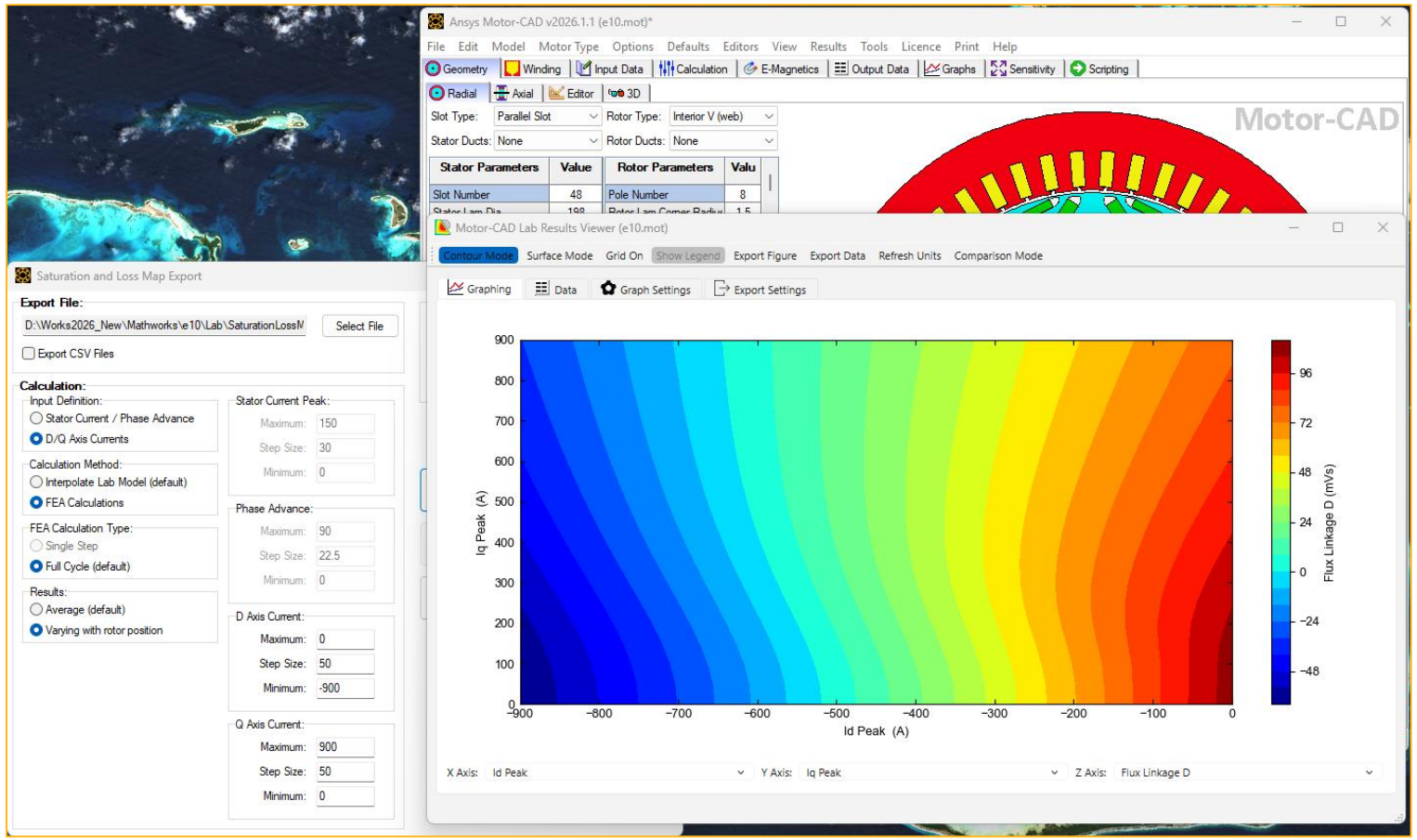


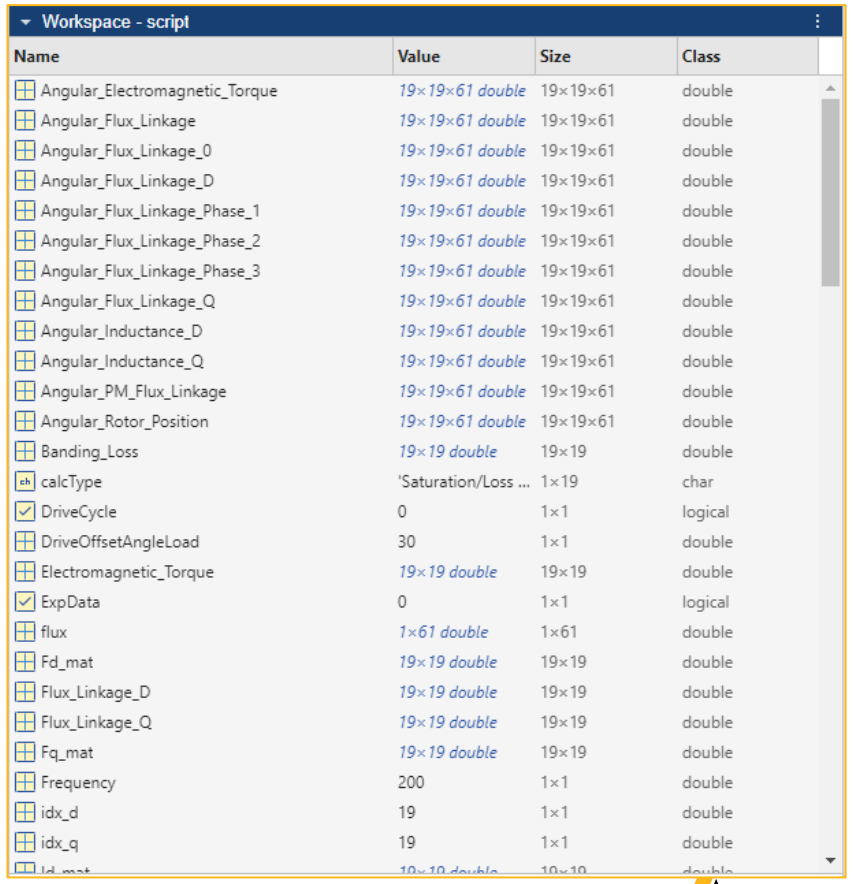
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Saturation and Loss Map Export (3)

- The output is automatically a .mat file with all the maps saved.





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